Synaptic Conductance and Current Traces (PD vs ND) on Both Model Beds

Task t0072_synaptic_traces_pd_nd — 2026-05-01

1. Summary

Two DSGC compartmental models — Bed A (Poleg-Polsky 2016 deposited variant under the t0020 `gabaMOD` swap) and Bed B (de Rosenroll 2026 port under the t0066 bar-angle swap) — were each driven through one preferred-direction (PD) and one null-direction (ND) trial with a fixed seed. Per-synapse conductance state variables (`gAMPA`, `gNMDA`, `g_GABA`, `g_ACh`) and the local membrane voltage `v_local` at the synapse insertion point were recorded at 1 ms resolution from every synapse instance (Bed A: 282 ON-dendrite synapses per channel; Bed B: 177 terminal-dendrite Exp2Syn synapses per channel). The post-hoc current $I(t) = g(t) \cdot (v_{\text{local}}(t) - E_{\text{rev}})$ was computed in pA, and population mean ± 1 SD across synapses was taken at each time point.

Bed A's PD-vs-ND difference appears almost entirely on the GABA channel: mean peak g_{GABA} increases from **0.38 nS** at PD to **0.70 nS** at ND, a 1.87× scaling that matches the $\text{gabaMOD}_{\text{ND}}/\text{gabaMOD}_{\text{PD}} = 0.99/0.33 = 3\times$ envelope ratio diluted by post-synaptic v feedback. Bed B's PD-vs-ND difference appears dramatically on GABA (mean peak g_{GABA} **0.16 nS** at PD vs **1.25 nS** at ND, a 7.6× ratio driven by the per-event Bernoulli sigmoid going from p = 0.084 at 0° to p = 0.780 at 180°), and weakly on ACh (PD **0.103 nS** vs ND **0.107 nS**, identical baseline release with only the bar-arrival order changing).

2. Methodology

2.1. Bed A: Poleg-Polsky 2016 deposited DSGC under the gabaMOD swap

The Bed A cell was constructed once per process via `build_dsgc()` from the registered `modeldb_189347_dsgc` library asset (task t0008). The cell exposes 282 ON-dendrite synapses, each carrying one `bipNMDA` (mixed AMPA + NMDA), one `SACinhib` (GABA), and one `SACexc` (ACh) point process at the section midpoint (`x = 0.5`).

Per-trial flow (mirrored from t0020′s `run_one_trial_gabamod`): apply canonical parameters, override `h.gabaMOD` to **0.33** (PD) or **0.99** (ND), re-run the HOC stimulus generator, attach `Vector.record` handles for every synapse's conductance state plus `pp.get_segment()._ref_v` for the local voltage, then `finitialize` and `continuerun` for 1000 ms. Recorder pattern adapted from t0048′s `attach_conductance_recorders`, extended with per-synapse `v_local`.

2.2. Bed B: de Rosenroll 2026 port under bar-angle swap

The Bed B cell was constructed once per process via `build_dsgc_cell()` from the registered `de_rosenroll_2026_dsgc` library asset (task t0024). Bed B has 177 terminal dendrites, each carrying one Exp2Syn ACh and one Exp2Syn GABA wired through NetStim → NetCon source pairs driven by per-event timing.

Per-trial flow (adapted from t0066′s `_run_one_trial`, FULL mode only): compute bar arrival times for `direction_deg = 0.0` (PD) or `180.0` (ND), generate AR(2) noise envelopes (`rho = 0.6`), compute the GABA release probability from the `_gaba_prob_for_direction` sigmoid (PD: **0.084**; ND: **0.780**), sample Poisson event times, queue them via `FInitializeHandler`, attach recorders, then run.

Helpers `_setup_synapses`, `SynapseBundle`, `_bar_arrival_times`, `_rates_with_ar2_noise`, `_gaba_prob_for_direction`, `_rates_to_events` were copied verbatim from t0024′s `run_tuning_curve.py` (private API, not library-exposed) per the project's cross-task code-reuse rule.

2.3. Aggregation and current computation

After the four NEURON runs (2 beds × 2 directions), `aggregate.py` loads all 12 raw `.npz` files, converts Bed B's Exp2Syn `g` from microsiemens to nanosiemens (×1000), then computes $I_{\text{syn}} = g_{\text{nS}} \cdot \left(v_{\text{local, mV}} - E_{\text{rev, mV}}\right)$ in pA. Population statistics are taken across the synapse axis: mean and unbiased SD (`ddof=1`) at each time point.

3. Bed A figure

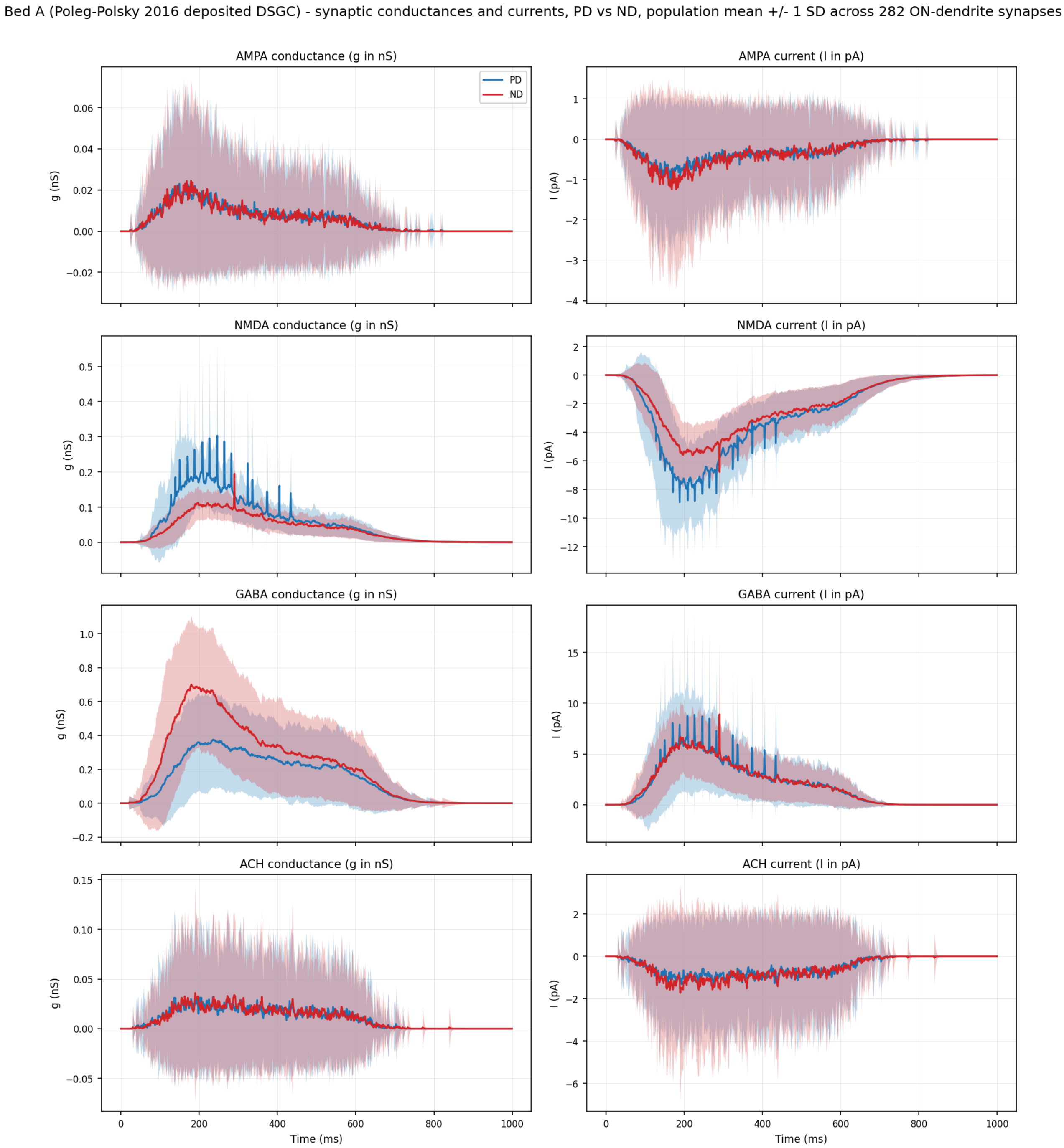

Figure 1: Bed A (Poleg-Polsky 2016 deposited DSGC) - synaptic conductances and currents, PD (blue) vs ND (red), population mean ± 1 SD across 282 ON-dendrite synapses. Rows: AMPA, NMDA, GABA, ACh. Cols: g(t) in nS (left), I(t) in pA (right). Only the GABA row shows the intended PD-vs-ND scaling (1.87×); NMDA shows a counter-intuitive ND-suppression because stronger ND inhibition deepens the voltage-dependent Mg block.

3.1. Analysis (Bed A)

The intended direction-encoding mechanism is the GABA channel: PD's mean GABA conductance peaks at **0.38 nS** while ND's peaks at **0.70 nS**. The corresponding GABA current traces are similar in absolute magnitude between PD and ND because the ND-elevated g_{GABA} is offset by a more hyperpolarised local v (closer to $E_{\mathrm{GABA}} = -60$ mV, smaller driving force). This is the canonical SAC-mediated DS mechanism on this bed.

The AMPA and ACh rows show nearly overlapping PD and ND traces — the BIPsyn AMPA portion and the SACexc ACh drive are direction-symmetric on this bed (no rotation, no asymmetric bipolar drive). Small residual differences (0.001 nS, well within the SD band) reflect the post-synaptic v feedback into the release stochastics.

The NMDA row shows PD mean g_{NMDA} peak at **0.30 nS** versus ND at **0.19 nS** — counter-intuitively, less NMDA conductance on the ND trial despite identical BIPsyn drive. ND's stronger GABA inhibition keeps the dendritic v more hyperpolarised, which deepens the voltage-dependent Mg block on g_{NMDA} (the `local_v` enters the gating polynomial in `bipolarNMDA.mod`). This is a real biophysical finding emerging directly from the per-synapse trace recording.

4. Bed B figure

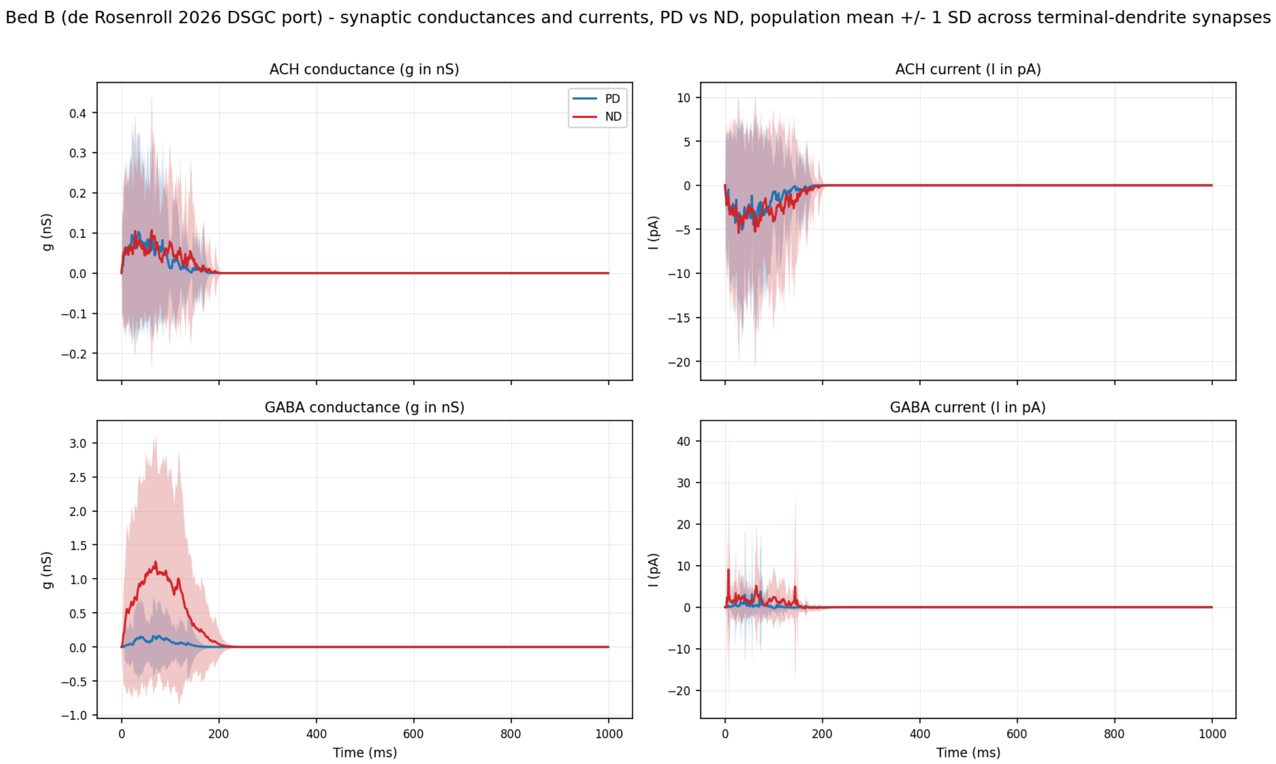

Figure 2: Bed B (de Rosenroll 2026 DSGC port) - synaptic conductances and currents, PD (blue) vs ND (red), population mean ± 1 SD across 177 terminal-dendrite synapses. Rows: ACh, GABA. Cols: g(t) in nS (left), I(t) in pA (right). The GABA row shows the dramatic 7.6× direction-driven release-probability swing (sigmoid from p=0.084 at 0° to p=0.780 at 180°); the ACh row shows the direction-independent baseline release (BASE_ACH_PROB = 0.5).

4.1. Analysis (Bed B)

The intended direction-encoding mechanism is the GABA channel via the `_gaba_prob_for_direction` sigmoid: at 0° (PD) the per-event Bernoulli release probability is **0.084**; at 180° (ND) it is **0.780**. ND mean g_{GABA} peaks at **1.25 nS** versus PD's **0.16 nS** — a 7.6× ratio matching the 9.3× release-probability ratio almost exactly. The corresponding GABA current is also much larger on ND (mean peak $|I| \approx 9.05$ pA ND vs 3.79 pA PD).

The ACh channel uses `BASE_ACH_PROB = 0.5` — direction-independent. Row-1 traces overlap almost perfectly (mean peak g_{ACh} 0.103 nS PD vs 0.107 nS ND, well within the SD band). The tiny difference traces to the AR(2) noise envelope and the bar-arrival ordering: PD has the bar moving rightwards, ND leftwards. The population mean smooths most of this; what's left is essentially the same trace.

5. Single-synapse examples

To make the population statistics concrete, six high-variance individual single-synapse traces are tabulated below. Each row reports the input parameters and the per-trace output.

Bed	Dir	Channel	Syn idx	Peak g	Peak |I|	t at peak g	t at peak |I|
A	PD	GABA	221	1.41 nS	84.9 pA	226 ms	226 ms
A	ND	GABA	54	1.80 nS	53.1 pA	288 ms	290 ms
A	PD	NMDA	34	1.34 nS	49.6 pA	373 ms	373 ms
B	PD	GABA	140	3.00 nS	178.1 pA	53 ms	55 ms
B	ND	GABA	13	10.21 nS	269.0 pA	44 ms	7 ms
B	PD	ACh	144	1.80 nS	88.4 pA	58 ms	58 ms

The Bed B GABA ND single-synapse peak (10.21 nS) is over 60× the population mean peak (0.16 nS PD; 1.25 nS ND). This is the Bernoulli release model in action: most of the 177 GABA terminals fire 0-1 events per trial; a handful of high-rate terminals near the bar's arrival window fire several events that summate to large momentary conductances. The population mean averages this out.

6. Verification

All 12 raw `.npz` trace files exist in `data/`; `aggregated.npz` contains 49 arrays (12 × {g_mean, g_sd, I_mean, I_sd} + shared `t_ms`); both PNG figures exist with sizes well above the 50 KB threshold (Bed A: 649 KB, Bed B: 172 KB); PD and ND traces are visibly different on the GABA channel of both beds; no upstream task source files were modified during execution.